

The Greenhouse Effect

Pupil sheet

Name: _____

Date: _____

Safety: No flame in the jar. The box is not waterproof: keep water and vinegar away from it.

1. What I see on the screen

On screen: **CO₂** (ppm), **Temperature** (°C) and **Illuminance** (lx), with charts of the last 30 minutes. I write the temperature down every 10 minutes. **Only the box** goes into the jar; the electronics stay outside.

Be careful what the experiment proves. A closed jar under a lamp warms up *mainly because it has a lid*: the lid stops the warm air leaving. To say anything about CO₂ I need a **control** — a second, identical jar, treated identically, with no extra gas.

Scatter of the rig (empty jars, 15 min) = _____ °C

A difference smaller than that *cannot be told apart*.

2. What I do on my own

First I predict — about the *cause*, not the result. The jar will warm up mainly because of:

☐ the lid (it stops convection) ☐ the walls (they absorb light) ☐ the CO₂ (it absorbs and re-emits infrared)

because _____

I check the symmetry before starting. Light: jar A _____ lx, jar B _____ lx. Do they differ by under 5%? ☐ yes ☐ no

Experiment 1 — the lid (lesson 1). Both jars with ordinary air; one with a lid, one without.

Minute	With lid (°C)	No lid (°C)	Difference
0			
10			
20			

Verdict (circle one): detected / borderline / not detected

Experiment 2 — CO₂ (lesson 2). Both with lids. **I make the same gesture in both:** I pour CO₂ from a bottle into one and air from an empty bottle into the other. I wait 10 minutes with the lamp off, then switch it on (minute 0).

CO₂: the gas jar _____ ppm

the control _____ ppm

Minute	With (°C)	CO ₂	Control (°C)	Difference
0				
10				
20				
25				

Verdict (circle one): detected / borderline / not detected

3. The conclusion I write

I predicted that, and the data show that _____

A garden greenhouse and the atmosphere warm differently, because _____

The very fact that the sensor can measure CO₂ proves that _____

Exit ticket

What warmed the jar, and how do you know?
